docked with an Agena vehicle achieving the first successful docking in space. And even with their success, all was not well. The joined spacecraft began to tumble end over end. The pilot managed to control the craft by activating thrusters intended for use during reentry. It worked, and the astronauts achieved the first successful docking in space, but it had been a close call.

A smoother docking was achieved by Gemini 10 in July 1966, crewed by John Young and Michael Collins. They not only rendezvoused and docked with an Agena vehicle, but also carried out maneuvers with the two spacecraft joined together.

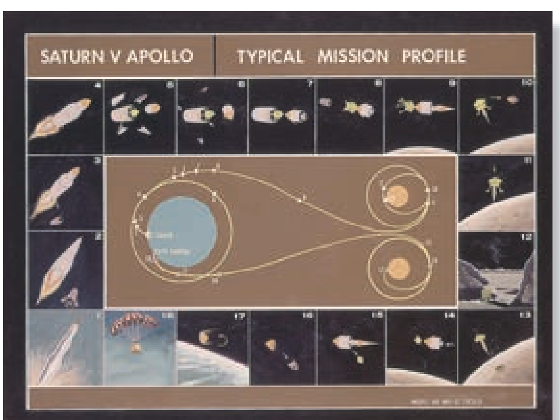
Space walking also ran into problems after the successful start with Gemini 4. During the Gemini 9 mission, astronaut Gene Cernan experienced severe difficulty when trying to work outside the spacecraft and had to curtail his EVA. But all difficulties were triumphantly overcome during the last Gemini mission, Gemini 12, in November 1966, when Edwin “Buzz” Aldrin carried out five hours of space walks without a problem.

ILL-FATED CREW
The crew of the Apollo 1 mission, Gus Grissom, Ed White, and Roger Chaffee, were the first casualties of the American space program. They died when fire broke out in their command module (left) during a routine ground test in January 1967.

Apollo gets under way

With the successful completion of the Gemini program, the path to the Moon was open. NASA had long decided on the best route to follow. Back in 1959, Wernher von Braun had chosen the name Saturn for a new series of powerful launchers. Like all his generation of rocket pioneers, von Braun had always assumed that a launcher of such power would be required for a manned Moon landing, because the space vehicle would have to be large and packed with heavy propellant to blast it off the Moon’s surface for the return to Earth.

When President Kennedy announced his time-scale for the Moon program, however, von Braun was clear that developing Nova would take too long. He suggested instead a plan involving the use of two smaller Saturn launchers. One would lift the crew into Earth orbit, while the other would lift the propellant. They would link up in orbit and proceed to the Moon. But this was not the plan adopted. Instead, in 1962, NASA opted for a “lunar orbit rendezvous” plan. This involved using a spacecraft consisting of a command and service module (CSM) and a lunar module (LM). Only the lunar module would land on the Moon’s surface, later blasting off to rejoin the CSM in lunar orbit before the return journey to Earth. The advantage of this plan was that relatively little propellant would be needed, allowing the spacecraft to be smaller and lighter, and thus lifted by a less powerful launcher.

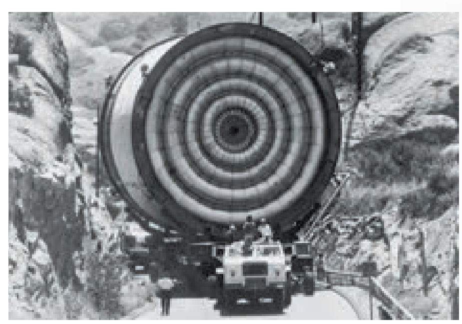


MOON MISSION PROFILE
This depiction of the various stages of the planned Apollo Saturn mission gives a graphic representation of the complexity of the operation. The key to the mission plan was the use of a lunar module to land on the surface of the Moon and then blast off to meet up with the command module in lunar orbit.

The scale of the effort required to put this plan into practice was awesome. While von Braun’s Marshall Space Center co-ordinated the work of literally hundreds of contractors and subcontractors developing the Saturn series of launchers, Grumman took on building a lunar module and North American wrestled with the problems posed by the command module. NASA’s additional facility, the 1961 Manned Spacecraft Center in Houston, Texas, today, the Johnson Space Center, took over as the nerve center for America’s space flights from 1965. By the mid-1960s NASA’s budget had risen to over $5 billion a year and around half a million people in the United States were carrying out work related to the space program, about 36,000 of them being employed by NASA.

Yet a poor contractor, North American, and a lack of attention by NASA to the problem of filling the spacecraft with pure oxygen making the interior atmosphere at normal pressure highly combustible led to corner-cutting and a lack of adequate attention to safety. Astronauts Gus Grissom, Ed White, and John Chaffee paid the price. On January 27, 1967, they were victims of a flash fire in the pressurized pure-oxygen atmosphere of their command module during a ground test in preparation for the first manned Apollo flight. A shocked NASA put a temporary stop to manned missions and embarked on a scrutiny of the Apollo spacecraft that led to over a thousand design changes being made.

Yet 1967 ended on a high for the space agency after the successful first test of the Saturn V launcher. This astonishing machine, the crowning glory of von Braun’s career, consisted of three



SATURN TRANSPORTATION
Transporting the three stages of the giant Saturn V rocket, from the different factories, ultimately reaching their destination – Cape Canaveral (then Cape Kennedy), was an achievement in its own right. The second stage was 33ft (10m) in diameter and had to be carried on a special vehicle to a barge for transportation by sea.